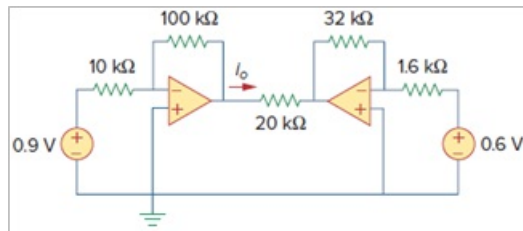


Problem

Find i_o in the op amp circuit of Fig. 5.100.

Figure 5.100:



Step-by-step solution

Step 1 of 5

Draw the circuit diagram as shown in Figure 1.

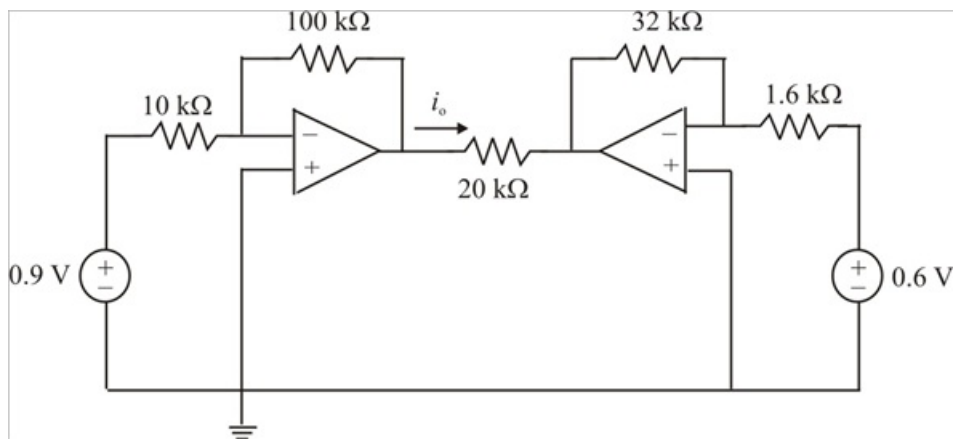


Figure 1

Step 2 of 5

The circuit is labeled as shown in Figure 2.

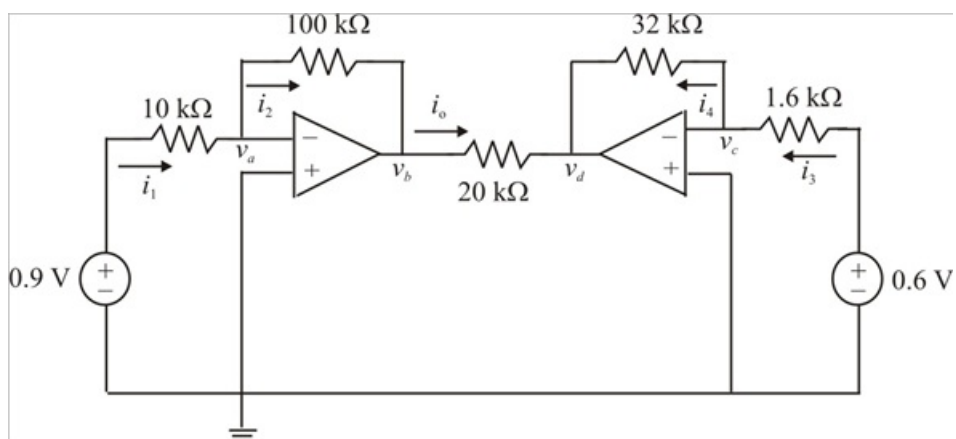


Figure 2

Step 3 of 5

Applying Kirchhoff's current law to node v_a ,

$$i_1 = i_2$$

$$\frac{0.9 - v_a}{10} = \frac{v_a - v_b}{100}$$

$$9 - 10v_a = v_a - v_b$$

$$11v_a - v_b = 9$$

$$\text{Therefore } 11v_a - v_b = 9 \dots\dots (1)$$

But $v_a = 0$ since non-inverting terminal is grounded. Then equation (1) becomes,

$$11v_a - v_b = 9$$

$$11(0) - v_b = 9$$

$$v_b = -9 \text{ V}$$

$$\text{Therefore } v_b = -9 \text{ V} \dots\dots (2)$$

Step 4 of 5

Applying Kirchhoff's current law to node v_c ,

$$i_3 = i_4$$

$$\frac{0.6 - v_c}{1.6} = \frac{v_c - v_d}{32}$$

$$12 - 20v_c = v_c - v_d$$

$$21v_c - v_d = 12$$

$$\text{Therefore } 21v_c - v_d = 12 \dots\dots (3)$$

But $v_c = 0$ since non-inverting terminal is grounded. Then equation (3) becomes,

$$21v_c - v_d = 12$$

$$21(0) - v_d = 12$$

$$v_d = -12 \text{ V}$$

$$\text{Therefore } v_d = -12 \text{ V} \dots\dots (4)$$

Step 5 of 5

But the current i_o can be calculated as:

$$\begin{aligned} i_o &= \frac{v_b - v_d}{20 \times 10^3} \\ &= \frac{-9 - (-12)}{20 \times 10^3} \\ &= \frac{3}{20 \times 10^3} \\ &= 150 \mu\text{A} \end{aligned}$$

Therefore the current i_o is $150 \mu\text{A}$.